

Evaluation of skin as a matrix for the detection and potential quantification of α -synuclein seeds in subjects with Parkinson's disease. Part 1.

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Summary

The purpose of this part of the study was to evaluate the performance of the first prototype α -synuclein seed amplification assay developed by Amprion to detect α -synuclein seeds in skin (skin_synSAA R&D-v1). The second part of the study includes an additional round of optimization, followed by another performance evaluation, and if successful, a final part looking at serial dilutions to quantify levels of syn-seeds in skin.

Methods

Homogenization of skin biopsies

Skin biopsies (n=93) from prodromal, sporadic Parkinson's disease (PD), and control individuals from the PPMI cohort were homogenized at Indiana University following the protocol established by Amprion. Briefly, each sample was homogenized to 1% w/v in 1X PBS (Cytiva) supplemented with 1X protease inhibitor (Roche). Homogenization was performed using a Precellys 24 Touch mechanical homogenizer and six 2.8mm stainless steel beads (Bertin Technologies). A total of five 45-second cycles at 6,800rpm were performed per sample. Crude skin homogenates were clarified by centrifuging 10 minutes at 10,000 x g (4°C), and the resulting supernatants were transferred into new tubes without collecting any debris. The supernatants were then mixed thoroughly, aliquoted, snap-frozen in liquid nitrogen, and stored at -80°C as 1% skin homogenate.

Skin_synSAA R&D-v1.

The skin_synSAA reaction mixture was the same as recently reported for cerebrospinal fluid (CSF)(Ma et al., 2024). Briefly, the amplification reaction was performed with two 2.45mm glass beads (Sigmund Lindner) in a final reaction volume of 100 μ L, consisting of 100mM PIPES pH6.50, 440mM NaCl, 10 μ M ThT, 0.1% sarkosyl, 0.3mg/mL recombinant α -synuclein (Amprion), and 40 μ L of 1% skin homogenate. Plates were incubated at 42°C with orbital shaking (800rpm) for 1min in a 15min cycle. Fluorescence was read every cycle (15min) using FLUOstar Omega (BMG). Each FLUOstar Omega was calibrated to ensure consistent and comparable measurements with the use of fluorescent dye.

Each sample was analyzed in triplicates and the maximum fluorescence (440nm/490nm) was used to determine results when applying a threshold of 6,000 relative fluorescence unit (RFU). A sample was defined as α Syn-SAA positive when all three replicates reached the threshold. If only two replicates reached the threshold, the sample was deemed inconclusive. Otherwise, the sample was deemed negative. Results were determined for both 24h and 48h, and the kinetic parameters were calculated as previously described for project 237 and averaged per sample.



References.

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